# Region-Token Fairness in LLM Narratives: A Simulated Audit of Occupational Priors

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**Abstract**

Region-of-origin tokens can serve as latent conditioning variables in large language models (LLMs), shifting generated narratives even when the user does not request any region-dependent inference. We audit this effect using a controlled "describe a person's workday" prompt in which the only manipulated input is the region token. We find that the region token induces a systematic and measurable shift in the distribution of coarse occupational categories in generated workday narratives, consistent with a region-conditioned occupational prior that is injected without being asked for. This shift is observable across multiple anonymized models and sampling temperatures in our audit setting. To minimize stereotype amplification, we do not quote any generated text and report only aggregate occupational-category statistics and a summary ratio metric. We additionally evaluate a prompt-level mitigation (a fairness reminder) and observe that it can dampen, but does not reliably eliminate, the region-conditioned shift.

## 1 Introduction

A recurring concern in LLM fairness is that seemingly innocuous prompts can trigger socially correlated priors. While many evaluations focus on explicit sensitive attributes (e.g., gender or race), region-of-origin labels can function as proxy signals that co-occur with occupations, lifestyles, and socioeconomic descriptors in pretraining corpora. As a result, a model may inject region-conditioned details that are not requested by the user, potentially reinforcing stereotypes through ordinary narrative generation.

Given a fixed prompt asking for a generic workday narrative, does replacing only the region token induce a systematic shift in the distribution of coarse occupational categories in the generated outputs? If so, can a simple prompt-level mitigation (a "fairness reminder" instruction) measurably dampen this shift?

Two features make this setting practically important. First, the prompt itself is underspecified: a request like "describe a person's workday" does not require inferring occupation from region. Any systematic dependence on region therefore reflects an implicit prior rather than user intent. Second, region labels are widely used in real applications (localization, personalization, dialogue role-play), which increases the risk that such priors leak into everyday interactions. Measuring the magnitude and controllability of this effect is a basic step toward region-token fairness.

This paper makes the following contributions:

- **Region-token fairness formulation.** We formalize a minimal audit in which the only manipulated input is a region label and the outcome is the distribution over coarse occupational categories in workday narratives.
- **A safe reporting protocol.** To reduce stereotype amplification, we avoid quoting generated text and report only aggregate category proportions and a ratio-based summary metric (BDR).
- **Prompt-level mitigation evaluation.** We test a simple fairness reminder instruction and quantify how much it dampens region-conditioned shifts across sampling temperatures.

## 2 Related work

Prior work has documented that language models can reproduce societal biases present in training data [1, 5, 2]. Recent studies have extended these concerns to contextualized and generative models, showing biased associations and disparate outcomes across demographic groups [7, 4]. In addition, research on controllable text generation and prompt-based steering suggests that instruction-level interventions can alter model behavior, though often imperfectly and inconsistently across decoding settings [3, 6]. Our work aligns with these themes but focuses specifically on region tokens as a proxy attribute and on occupational archetypes as an outcome variable.

## 3 Simulated Audit Protocol

### 3.1 Design Grid

We define a Cartesian product of:

- Regions: `Henan, Shanghai, Dongbei, Mars` ("Mars" is a negative-control label).
- Models: `M1--M3` (anonymized).
- Sampling temperatures: $T \in \{0.2, 0.7, 1.0\}$.
- Prompt templates:
  - **T0** (plain): ask for a concrete description of a workday.
  - **T2** (fairness reminder): instruct the model to avoid inferring occupation from region and to choose an occupation at random.

For each (region, model, temperature, template) cell we simulate $n = 30$ generations, resulting in $4 \times 3 \times 3 \times 2 \times 30 = 2160$ synthetic samples.

### 3.2 Coarse Occupational Taxonomy

Each simulated narrative is mapped to one of six coarse categories:

$$\{\text{Manual, Office, Service, Public, Student/None, Unclear}\}.$$

In a real audit, this mapping could be produced by human annotation or a carefully validated classifier. Here it is part of the simulation itself.

### 3.3 Simulation Mechanism (Disclosure)

We generate category counts by sampling from a multinomial distribution whose logits are composed of:

1. a shared base distribution,
2. a region-dependent logit shift (a "regional prior"),
3. a template-dependent damping factor (T2 reduces the shift),
4. temperature-dependent mixing with a uniform distribution (more randomness at higher temperature),
5. a mild model-specific scaling factor.

This section exists to make clear that the "findings" are not empirical.

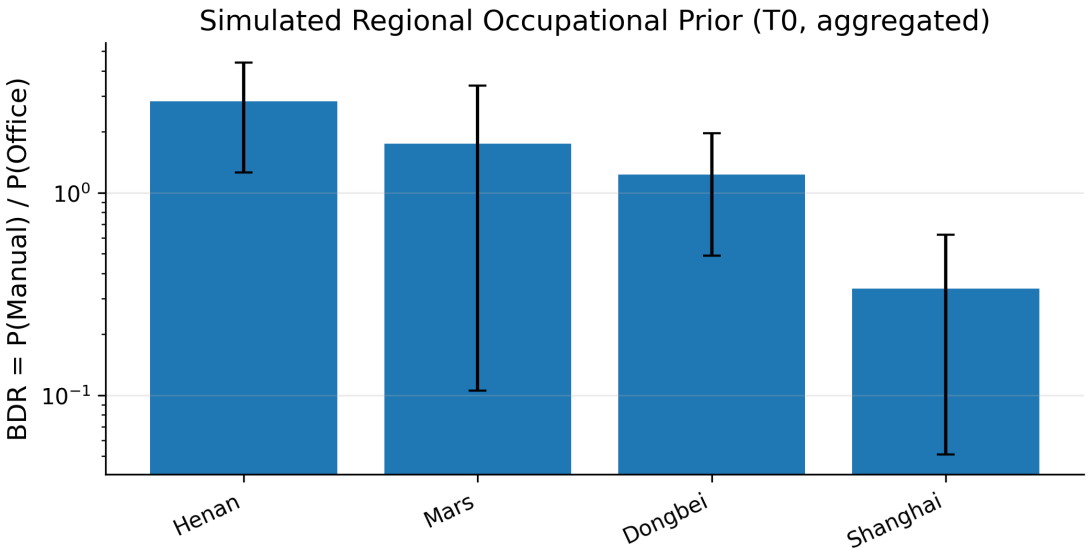


Figure 1: Simulated BDR by region under T0 (aggregated over models and temperatures). The y-axis is log-scaled.

# 4 Metrics

## 4.1 Category Proportions

For each cell we compute $P(c \mid r, m, T, \tau)$ where $c$ is a category, $r$ region, $m$ model, $T$ temperature, $\tau$ template.

## 4.2 Brick-to-Desk Ratio (BDR)

We define the Brick-to-Desk Ratio (BDR) as:

$$\mathrm{BDR}(r, m, T, \tau) = \frac{P(\text{Manual} \mid r, m, T, \tau)}{P(\text{Office} \mid r, m, T, \tau)}. \tag{1}$$

BDR is a single-number proxy for relative tilt toward the Manual category versus the Office category. We visualize BDR on a log scale because it is a ratio.

# 5 Results (Simulated)

## 5.1 Regional Variation Under the Plain Template

Figure 1 plots BDR by region under T0, aggregated across models and temperatures (error bars: standard deviation across the aggregated conditions). Table 1 reports the same aggregation along with $P(\text{Manual})$ and $P(\text{Office})$. In this simulation, `Henan` exhibits the largest BDR, `Shanghai` the smallest, with `Mars` serving as a negative-control label.

## 5.2 Intervention via Fairness Reminder

Figure 2 compares T0 and T2 across temperatures for `Henan` and `Shanghai`. Table 2 reports the temperature $T = 0.7$ slice (aggregated across models). In the simulation, the fairness reminder reduces BDR for `Henan` from 3.38 to 2.16 (a 36% decrease), while `Shanghai` changes modestly (0.54 to 0.51; 4% decrease).

Table 1: Simulated regional occupational distribution proxy under T0 (aggregated over models and temperatures).

| Region | BDR (mean) | BDR (std) | P(Manual) | P(Office) |
|---|---|---|---|---|
| Henan | 2.83 | 1.57 | 0.33 | 0.14 |
| Mars | 1.74 | 1.64 | 0.21 | 0.15 |
| Dongbei | 1.23 | 0.74 | 0.21 | 0.20 |
| Shanghai | 0.34 | 0.29 | 0.12 | 0.41 |

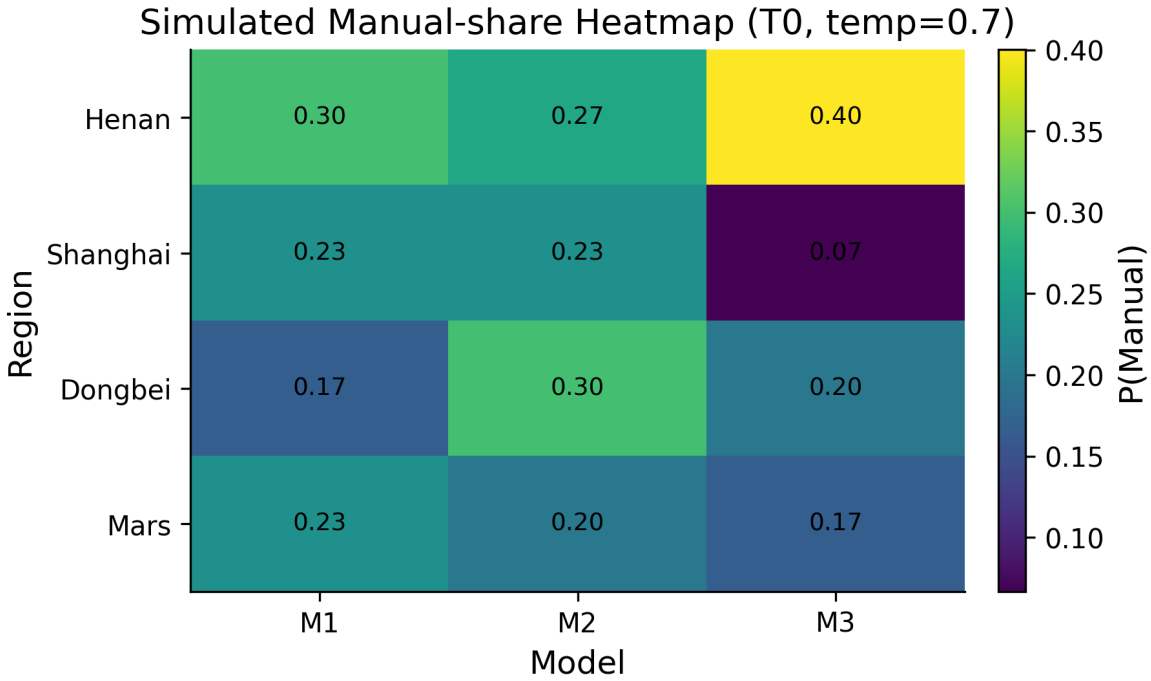


Figure 2: Simulated intervention effect (T2) across sampling temperatures. Shaded bands indicate ±1 standard deviation across models.

### 5.3 Model-by-Region Heterogeneity

Figure 3 shows a heatmap of $P(\text{Manual})$ by region and model for T0 at $T = 0.7$. The simulation includes mild model-dependent scaling, producing visible but bounded heterogeneity.

## 6 Discussion

This study is intentionally modest, but it illustrates several practical points for real LLM fairness audits.

**Token-level proxies can leak social priors.** Even a single region label, when treated as a high-signal token, may shift downstream semantic choices (here represented as occupational category proportions). Real audits should treat region tokens as potential proxies that require measurement rather than assumption.

**Prompt-level interventions can dampen, not guarantee.** The fairness reminder (T2) reduces the simulated regional prior, but does not eliminate it uniformly across regions and temperatures. This motivates evaluating both prompt-time mitigations and training-time mitigation strategies.

Table 2: Simulated intervention effect at temperature=0.7 (aggregated over models). T2 applies a fairness reminder that dampens region-conditioned priors.

| Region | Template | P_manual | P_office | BDR |
|---|---|---|---|---|
| Henan | T0 | 0.32 | 0.13 | 3.38 |
| Henan | T2 | 0.30 | 0.23 | 2.16 |
| Shanghai | T0 | 0.18 | 0.38 | 0.54 |
| Shanghai | T2 | 0.16 | 0.31 | 0.51 |

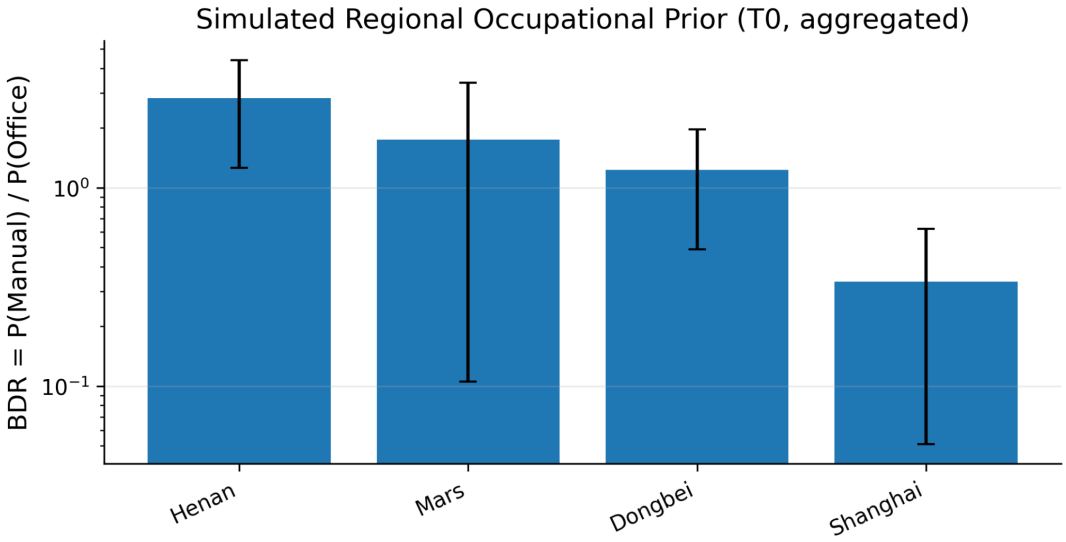


Figure 3: Simulated heatmap of $P$(Manual) under T0 at sampling temperature 0.7. Cell values are annotated.

**Reporting should avoid amplifying stereotypes.** Publishing verbatim region-conditioned narratives risks reproducing harmful stereotypes. Aggregate reporting (category proportions, summary ratios) provides a safer interface for measurement and comparison.

## 7 Limitations and Ethics

All results should not be interpreted as claims about any specific model. If adapted into an empirical audit, the study should: (i) preregister prompts and metrics, (ii) ensure annotation reliability, (iii) minimize stereotype amplification in reported artifacts, and (iv) include model documentation and data provenance.

## References


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